

BLUECARBON AI PLATFORM

Master Technical Specification & Complete Feature-by-Feature Operational Manual

Platform Overview: BlueCarbon AI is a full-stack, enterprise-grade Measurement, Reporting, and Verification (MRV) platform designed for coastal blue carbon ecosystems (mangroves, seagrasses, salt marshes) and adaptable to inland terrestrial forests. The system combines high-resolution GIS satellite multispectral imaging, IoT ocean buoy sensor streaming, AI explainable carbon & reversal risk prediction, and Web3 Polygon POS blockchain evidence ledgering. It provides tailored operational interfaces for Ministry of Earth Sciences (MoES) auditors, NGO project developers, and the general public.

Component / Module	Tech Stack & Libraries	Operational Functionality
Top Navigation Bar	React + Lucide Icons + TailwindCSS	Role switching, site dropdown, live stream toggle, raw dataset exports & IST clock
GIS Digital Twin Map	Leaflet.js + Esri World Imagery + Carto Voyager	Real satellite map tiles, 4 spectral layer modes, pin popups & custom ROI drawing tool
IoT Sensor Stream	Digital Twin WebSocket/Timer + Recharts	6 sensor metric gauges, 30s live history LineChart & automated anomaly detection
AI Carbon Engine	IPCC Tier 3 Equations + SHAP Explainable AI	AGB/BGB/SOC pool breakdown BarChart, risk scoring (0-100%) & 2020-2026 forecast
Stress Simulator	Stochastic Hazard Engine + Smart Contract Triggers	Simulates Cyclone, Oil Spill, Deforestation & Heatwave hazards with carbon loss math
Web3 Evidence Ledger	CryptoJS SHA-256 + Canvas Confetti + Polygon POS	256-bit hashing, Merkle root proofs, searchable tx log & printable MRV certificate modal
MoES Verifier Portal	Temporal Change Differencing + Fraud Audit Check	2020 Baseline vs 2026 Present satellite differencing, double-counting audit & credit sign-off
Backend REST Server	Node.js + Express + CORS + FS Static Middleware	Serves REST APIs (/api/hotspots, /api/telemetry, /api/simulate-disaster) & production build

1. Exhaustive Tool & Feature Operational Breakdown (Part 1)

Header Navigation Toolbar (Navbar.jsx)

Tool / Feature Name	Detailed Operational Explanation
Brand Title Logo	Renders BlueCarbon AI brand icon with pulsing green status indicator.
Forest Hotspot Selector	Dropdown allowing users to switch between 5 coastal blue carbon ecosystems. Dynamically updates all maps, charts, sensors, and audit logs.
Project NGO Role Button	Activates DEVELOPER mode. Grants full access to custom polygon drawing, disaster stress testing, telemetry controls, and system parameters.
MoES Auditor Role Button	Activates AUDITOR mode and automatically navigates to the MoES Verifier Workflow tab for government verification and credit minting.
Public Registry Role Button	Activates PUBLIC transparency mode and navigates to the Web3 Evidence Ledger tab for open credit verification and public ledger inspection.
IoT Streaming Toggle Button	Toggles live telemetry streaming updates on or off. Pausing freezes gauge readings for static analysis.
Raw Datasets Export Menu	Provides direct download links for master GeoJSON dataset (JSON), IoT sensor logs (CSV), and IPCC biomass factors (CSV).
System Clock Display	Displays live Indian Standard Time (IST) clock updated every second.
6 Main Navigation Tabs	Tabs for GIS Digital Twin, IoT Ocean Buoy Stream, AI Carbon Engine, Disaster Simulator, Web3 Ledger, and MoES Verifier Portal.

GIS Digital Twin Map (DigitalTwinMap.jsx)

Tool / Feature Name	Detailed Operational Explanation
Leaflet Map Renderer	Loads Esri World Imagery satellite tiles and Carto labels centered on exact GPS coordinates of the selected forest site.
Hotspot Map Pins	Custom animated map markers for all 5 hotspots; clicking any marker switches the active forest location.
Active Hotspot Info Card	Displays site name, state, ecosystem type, area hectarage, NDVI, canopy height, and total carbon stock.
True Color Spectral Layer	Renders standard satellite natural color view.
NDVI Spectral Layer	Renders Normalized Difference Vegetation Index highlighting canopy vegetation health and leaf density.
NDWI Spectral Layer	Renders Normalized Difference Water Index highlighting surface water and leaf moisture content.
LiDAR Height Mesh Layer	Visualizes canopy height structure and 3D forest profile.
Draw Custom ROI Polygon Tool	Simulates drawing a custom region of interest boundary and calculates custom area hectarage, estimated carbon stock, average NDVI, and reversal risk.
Disaster Warning Banner	Displays animated emergency alert overlay across the map whenever a disaster event is active.

IoT Ocean Telemetry Stream (TelemetryStream.jsx)

Tool / Feature Name	Detailed Operational Explanation
Telemetry Stream Banner	Displays buoy node ID, stream frequency (2.5s), and IPFS mesh connectivity status.
Salinity Gauge	Displays real-time water salinity in parts per thousand (ppt) with baseline reference.

Soil Organic Carbon (SOC) Gauge	Displays soil organic carbon percentage down to 1-meter sediment depth.
Water Temperature Gauge	Monitors submerged water temperature probe readings in °C.
Sediment pH Gauge	Monitors soil alkalinity/acidity balance.
Sea Level Anomaly Gauge	Displays tidal gauge sea level anomaly in mm.
Dissolved Oxygen Gauge	Displays dissolved marine oxygen concentration in mg/L.
Live Telemetry Stream LineChart	Recharts graph rendering past 30-second live streaming history for salinity, temp, and SOC %.
Automated Anomaly Log Table	Logs high temp (>31°C), hypersaline (>40 ppt), or SOC drop (<2.5%) alerts automatically.
Dynamic Site Reset	Automatically resets gauge baselines and telemetry arrays when switching forest locations.

2. Exhaustive Tool & Feature Operational Breakdown (Part 2)

AI Carbon Engine (CarbonAiEngine.jsx)

Tool / Feature Name	Detailed Operational Explanation
Scientific Methodology Banner	Displays IPCC Tier 3 methodology standard, total baseline stock (k tCO ₂ e), and annual sequestration rate (t/ha/yr).
Carbon Pool Breakdown BarChart	Horizontal BarChart rendering Above-Ground Biomass (AGB), Below-Ground Biomass (BGB), and Soil Organic Carbon (SOC).
Quick Pool Metric Cards	Cards showing exact t/ha values for AGB, BGB, and SOC pools.
Carbon Reversal Risk Progress Bar	Visual progress bar displaying risk score (0-100%) and risk tier (LOW, MEDIUM, HIGH, CRITICAL).
SHAP Explainable AI Attribution	Breaks down feature drivers influencing reversal risk (coastal erosion, canopy trend, storm surge, salinity).
Sequestration Forecast AreaChart	Multi-year carbon stock trajectory graph from 2020 to 2026.

Environmental Stress Simulator (DisasterSimulator.jsx)

Tool / Feature Name	Detailed Operational Explanation
Category 4 Cyclone Card	Simulates 185 km/h winds, defoliation, 15% carbon loss, and triggers EMERGENCY_FREEZE_CYCLONE_DAMAGE.
Coastal Oil Spill Card	Simulates petroleum slick smothering roots, 8% carbon loss, and triggers ALERT_POLLUTION_SOIL_TOXICITY.
Illegal Deforestation Card	Simulates aquaculture clearing, 22% carbon loss, and triggers CRITICAL_FRAUD_ILLEGAL_CLEARING.
Marine Heatwave Card	Simulates sea temp >33.5°C, thermal bleaching, 5% carbon loss, and triggers WARNING_THERMAL_BLEACHING_RISK.
Reset Ecosystem Baseline Button	Restores ecosystem health parameters and clears active disaster state.
Active Disaster Response Panel	Displays digital twin reaction, AI risk score jump, and smart contract emission freeze status.

Web3 Evidence Ledger (BlockchainLedger.jsx)

Tool / Feature Name	Detailed Operational Explanation
Generate MRV Certificate Button	Triggers canvas-confetti animation and opens printable certificate modal.
SHA-256 Payload Hash Card	Displays 256-bit cryptographic hash, payload origin, and verification match status.
Merkle Root Proof Card	Displays tree height 4 Merkle root proof hash derived from 4,096 telemetry blocks.
Blockchain Network Stats Card	Displays Polygon POS block height #19,842,105, block time (2.1s), ERC-1155 standard & state tamper check.
Smart Contract Log Table	Searchable table filtering block, timestamp, event, txHash, signer, and gas used.
Printable MRV Certificate Modal	Displays project details, total verified carbon, SHA-256 hash, QR code, and Print/Save PDF action (window.print()).

MoES Verifier Workflow (AuditorWorkflow.jsx)

Tool / Feature Name	Detailed Operational Explanation
Temporal Satellite Differencing	Side-by-side comparison of 2020 Baseline vs 2026 Present satellite imagery with active growth calculation and vegetation expansion %.

Automated AI Fraud Checks	Validates double-counting prevention against Verra & Gold Standard, spectral match, and permanence threshold.
Official Assessment Notes Textarea	Allows government auditors to enter field inspection comments, saved independently per forest location.
Approve & Mint Carbon Credits Button	Triggers confetti, records digital sign-off, and marks claim APPROVED.
Reject Claim Button	Marks claim REJECTED.

Host Server & Backend REST APIs (server.js)

Tool / Feature Name	Detailed Operational Explanation
GET /api/health	Returns system online status, platform title, uptime, and port.
GET /api/hotspots	Returns master GeoJSON dataset for all 5 blue carbon hotspots.
GET /api/telemetry	Returns live ocean buoy sensor stream data.
POST /api/simulate-disaster	Executes disaster hazard calculations and returns blockchain event codes.
POST /api/verify-mrv	Hashes MRV payloads using SHA-256 and returns Merkle root proofs.

3. Complete Technical & Scientific Glossary of Terms & Definitions

Term / Concept	Scientific & Technical Definition
Blue Carbon	Carbon captured and stored by coastal and marine ecosystems, primarily mangroves, seagrass meadows, and tidal salt marshes.
MRV (Measurement, Reporting & Verification)	A multi-step scientific and auditing framework used to measure carbon stocks, report findings, and verify credit claims before issuance.
GIS (Geographic Information System)	A framework for gathering, managing, and analyzing spatial satellite and geographic data linked to exact Earth coordinates.
Digital Twin	A dynamic virtual replica of a physical real-world ecosystem updated continuously using live IoT sensor streams and satellite data.
NDVI (Normalized Difference Vegetation Index)	A multispectral satellite index $(NIR - Red)/(NIR + Red)$ measuring green vegetation density, chlorophyll absorption, and canopy health.
NDWI (Normalized Difference Water Index)	A satellite index $(Green - NIR)/(Green + NIR)$ used to monitor surface water extent, leaf moisture content, and inundation.
LiDAR (Light Detection and Ranging)	An optical remote sensing technology using pulsed laser light to measure exact 3D vegetation canopy height and vertical forest structure.
AGB (Above-Ground Biomass)	All living plant matter located above the soil surface, including tree trunks, branches, bark, and foliage (measured in t/ha).
BGB (Below-Ground Biomass)	All living root biomass located beneath the soil surface that anchors trees and transfers carbon into sediments (measured in t/ha).
SOC (Soil Organic Carbon)	Carbon stored in organic matter within coastal muddy sediment layers down to 1-meter depth (representing 70-80% of total blue carbon).
tCO₂e (Metric Tons of CO₂ Equivalent)	The standard universal unit for measuring greenhouse gas emissions and carbon sequestration offset credits.
IPCC Tier-3 Methodology	The highest accuracy level defined by the Intergovernmental Panel on Climate Change utilizing high-resolution spatial mapping and species-specific allometric models.
SHAP (SHapley Additive exPlanations)	A game-theoretic Explainable AI (XAI) technique that quantifies the exact percentage contribution of individual climate drivers to risk predictions.
Reversal Risk	The probability (0-100%) that stored carbon will be accidentally released back into the atmosphere due to natural disasters (cyclones, heatwaves) or human clearing.
SHA-256 Cryptographic Hash	A 256-bit cryptographic hash algorithm that transforms data into an irreversible 64-character string, guaranteeing data tamper-proofing.
Merkle Root Proof	A cryptographic hash tree where every leaf node represents a telemetry block, allowing large datasets to be verified with a single root hash on-chain.
Polygon POS Blockchain	A high-speed, eco-friendly Layer-2 Ethereum-compatible blockchain network used for immutable audit logs and carbon credit smart contracts.
ERC-1155 Token Standard	A multi-token blockchain standard supporting environmental carbon credits, fractional certification, and cryptographic minting.
Double-Counting Fraud	The fraudulent practice where the exact same carbon credit offset is claimed or sold twice across different registries.
MoES (Ministry of Earth Sciences)	The Indian government ministry governing oceanography, meteorology, atmospheric sciences, and coastal ecosystem monitoring.

ISRO (Indian Space Research Organisation)	India's national space agency providing multispectral Earth observation satellite imagery (Sentinel-2 / EOS series).
Salinity (ppt)	Concentration of dissolved salts in water measured in parts per thousand (ppt), critical for mangrove root respiration and growth.
Dissolved Oxygen (mg/L)	The volume of free oxygen dissolved in water essential for marine organisms and submerged seagrass photosynthesis.